

# **Bias/Variance and Ensembling**

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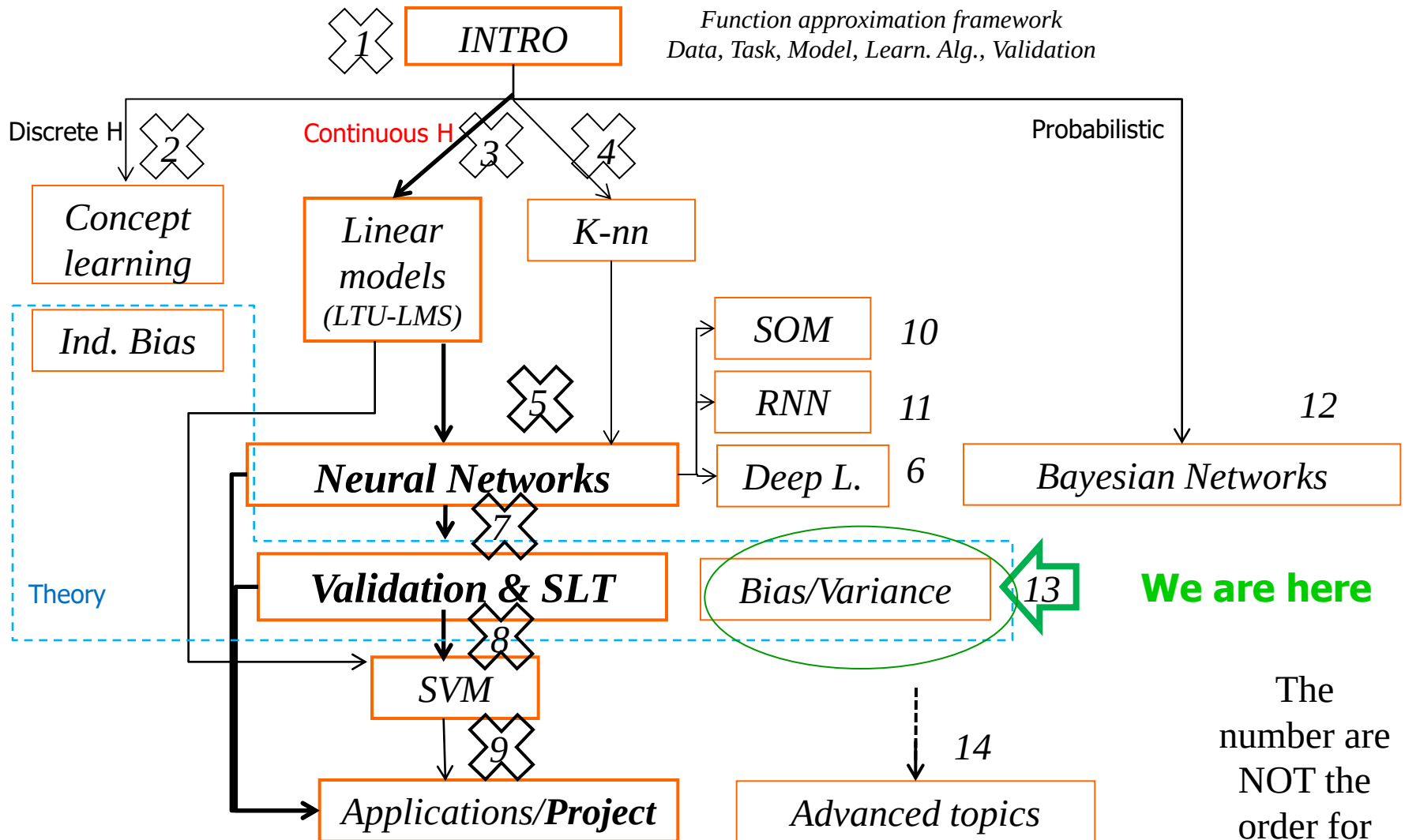
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# ML Course structure

## Where we are



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# Bias-Variance

- A training set is only one possible realization from the universe of data: different training set can provide different estimate
- Decomposition of expected (on various TR) error at a point  $x$ :
  - **Bias**: quantifies the discrepancy between true function and  $h(x)$  (averaged on data): if  $H$  is too small it is high
  - **Variance**: quantifies the variability of the response of model  $h$  for different realizations of the training data: due to high flexibility
  - **Noise**: because the label include random error
    - E.g. if for a given  $x$  there are more than one possible  $d$
- Assume the **regression** scenario with target  $y$  and  $L_2$  (*squared error loss*)

Credits: Pictures from Macskassy, Bishop, Bengio

# Bias-Var. Analysis in Regression



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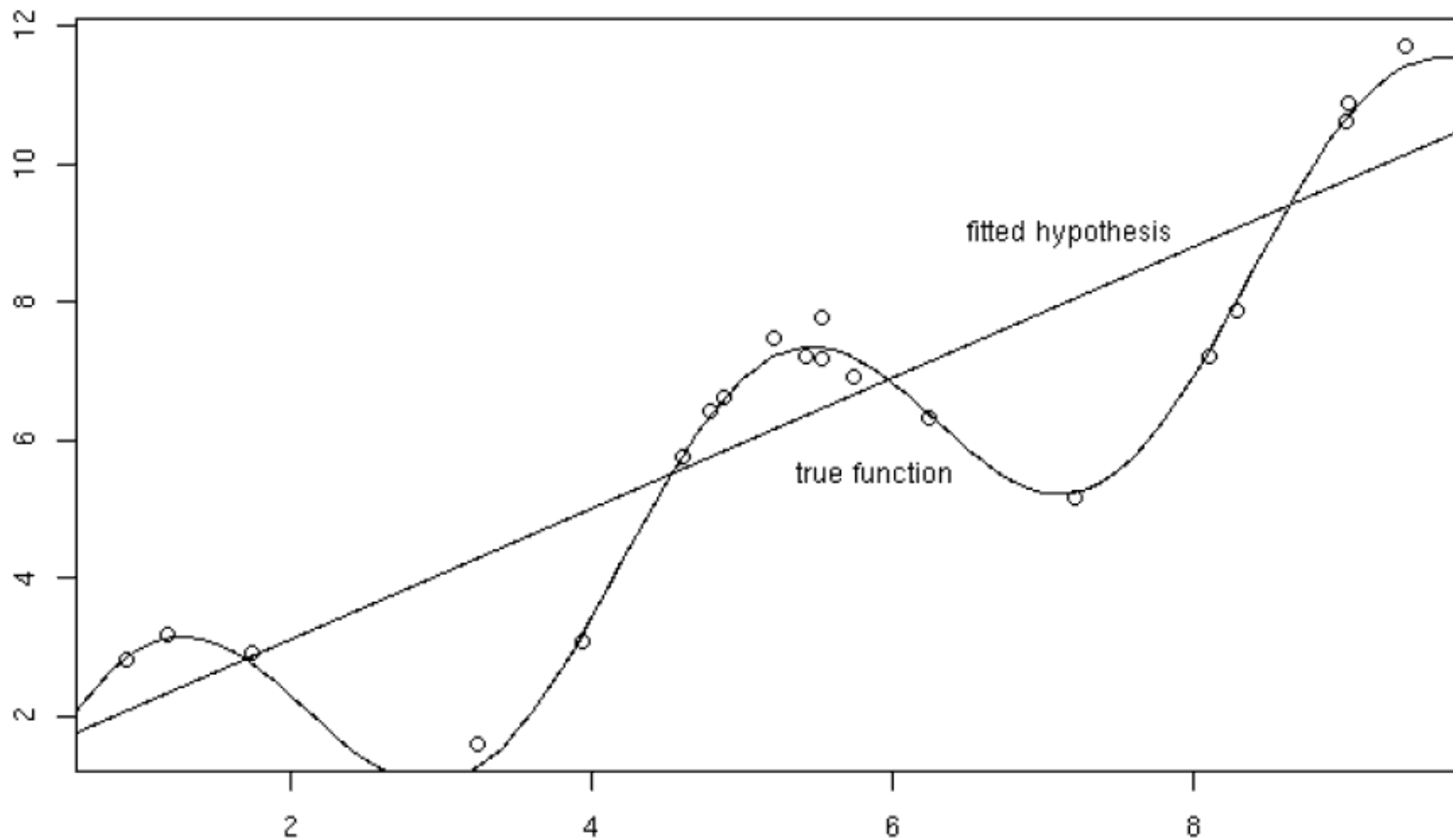
- Suppose we have examples  $\langle \mathbf{x}, y \rangle$  where the true function is  $y = f(\mathbf{x}) + \varepsilon$  and where  $\varepsilon$  is Gaussian noise with zero mean and standard deviation  $\sigma$ .
- In linear regression, given a set of examples  $\langle \mathbf{x}_i, y_i \rangle \quad i=1..l$  we fit a linear hypothesis  $h(\mathbf{x}) = \mathbf{w}\mathbf{x} + w_0$ , such as to minimize sum-squared error over the training data:

$$\sum_{i=1}^l (y_i - h(\mathbf{x}_i))^2$$

- 1) Because of the hypothesis class that we chose (linear hypotheses) for some function  $f$ , we will have a systematic prediction error
- 2) Depending on the data set we have, the parameters  $\mathbf{w}$  that we find will be different.

# Example: 20 points

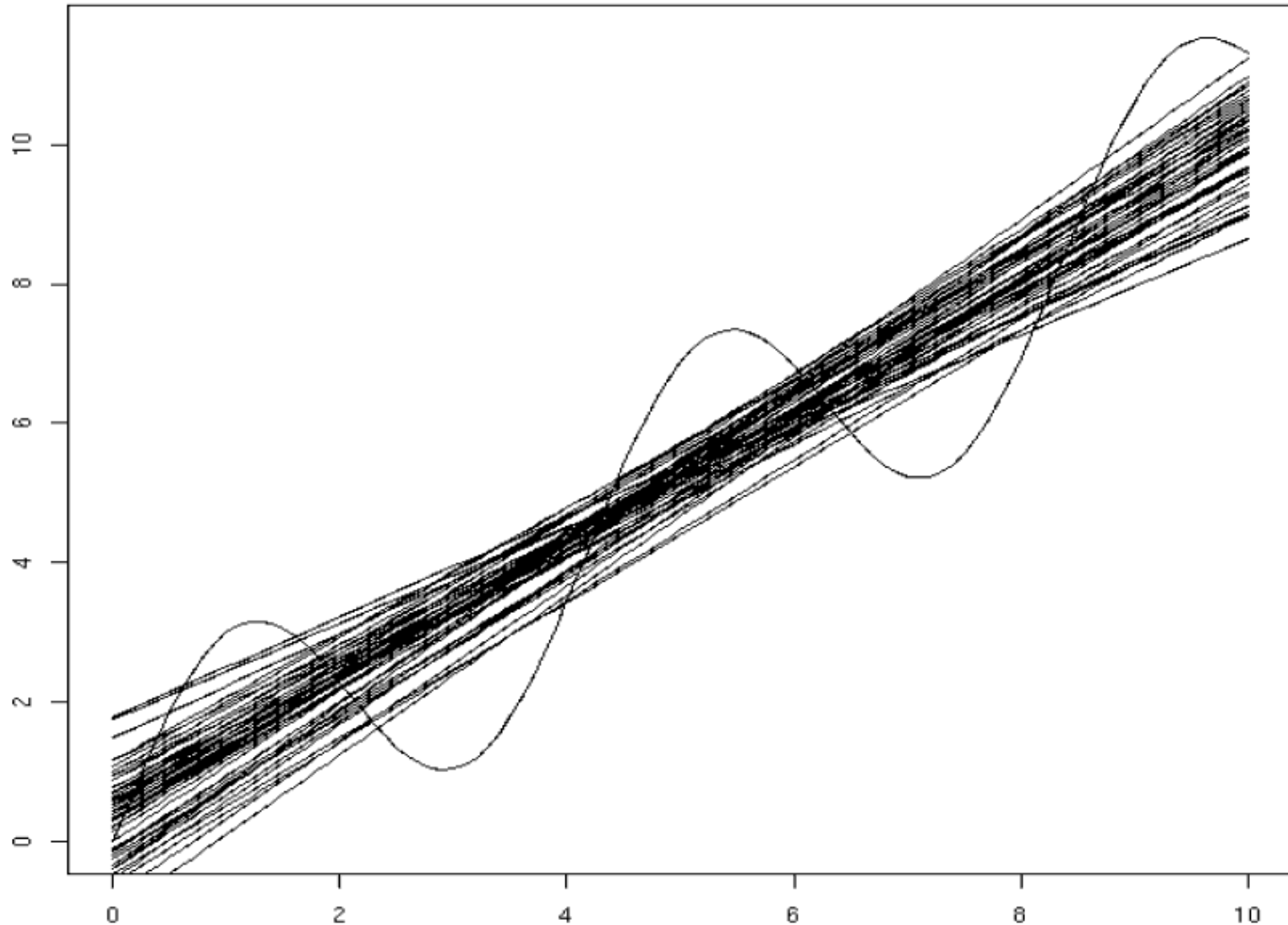
$$y = x + 2 \sin(1.5x) + N(0,0.2)$$



# Example: 50 fits (each on 20 points)



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# Bias-Var. Analysis

- Given a new data point  $\mathbf{x}$ , what is the **expected prediction error**?
- Assume that the data points are drawn i.i.d.\* from a unique underlying probability distribution  $P$
- The goal of the analysis is to compute, for an arbitrary new point  $\mathbf{x}$ ,

$$E_P \left[ (y - h(\mathbf{x}))^2 \right]$$

Note: there is a different  $h$  (and  $y$ ) for each different “extracted” training set :

where  $y$  is the value for  $\mathbf{x}$  that could be present in a data set, and the expectation is over all training sets drawn according to  $P$ .

- We will decompose this expectation into three components:  
bias, variance and noise

\* i.i.d: independent and identical distributed

# Recall of basic statistics

- Let  $Z$  be a random variable with possible values  $z_i$ ,  $i=1..l$  and with probability distribution  $P(Z)$
- The **expected value** or **mean** of  $Z$  is:

$$\bar{Z} = E_P[Z] = \sum_{i=1}^l z_i P(z_i)$$

- If  $Z$  is continuous the sum is replaced by an integral, and the distribution by a density function.
- The **variance** of  $Z$  is:

$$\begin{aligned} Var[Z] &= E[(Z - \bar{Z})^2] \\ &= E[Z^2] - \bar{Z}^2 \end{aligned}$$

(this last line is proven on the next slide)



# Variance Lemma

- $$\begin{aligned} Var[Z] &= E[(Z - \bar{Z})^2] = \sum_{i=1}^l (z_i - \bar{Z})^2 P(z_i) = \\ &= \sum_{i=1}^l (z_i^2 - 2z_i\bar{Z} + \bar{Z}^2) P(z_i) = \\ &= \sum_{i=1}^l z_i^2 P(z_i) - 2\bar{Z} \sum_{i=1}^l z_i P(z_i) + \bar{Z}^2 \sum_{i=1}^l P(z_i) = \\ &= E[Z^2] - 2\bar{Z}\bar{Z} + \bar{Z}^2 \cdot 1 = \\ &= E[Z^2] - \bar{Z}^2 = Var[Z] \end{aligned}$$

- We will use the form

$$E[Z^2] = \bar{Z}^2 + Var[Z]$$

# Bias-Var. Decomposition (1)

$$\begin{aligned} E_P [(y - h(\mathbf{x}))^2] &= E_P [h(\mathbf{x})^2 - 2yh(\mathbf{x}) + y^2] \\ &= E_P [h(\mathbf{x})^2] + E_P [y^2] - 2E_P[y]E_P[h(\mathbf{x})] \quad * \end{aligned}$$

- Let  $\bar{h}(\mathbf{x}) = E_P[h(\mathbf{x})]$  denote the mean prediction of the hypothesis at  $\mathbf{x}$ , when  $h$  is trained with data drawn from  $P$
- For the first term, using the variance lemma\*, we have:

$$E_P [h(\mathbf{x})^2] = E_P [(h(\mathbf{x}) - \bar{h}(\mathbf{x}))^2] + \bar{h}(\mathbf{x})^2$$

- Note that  $E_P[y] = E_P[f(\mathbf{x}) + \varepsilon] = f(\mathbf{x})$
- For the second term, using the variance lemma\*, we have:

$$E_P [y^2] = E_P [(y - f(\mathbf{x}))^2] + f(\mathbf{x})^2$$

\* Lemma:  $E[Z^2] = \text{Var}[Z] + \bar{Z}^2$

# Bias-Var. Decomposition (2)

- Putting everything together, we have:

$$\begin{aligned}
 E_P [(y - h(\mathbf{x}))^2] &= E_P [(h(\mathbf{x}) - \bar{h}(\mathbf{x}))^2] + \bar{h}(\mathbf{x})^2 - 2f(\mathbf{x})\bar{h}(\mathbf{x}) + f(\mathbf{x})^2 + E_P [(y - f(\mathbf{x}))^2] \\
 &= E_P [(h(\mathbf{x}) - \bar{h}(\mathbf{x}))^2] + (\bar{h}(\mathbf{x}) - f(\mathbf{x}))^2 + E_P [(y - f(\mathbf{x}))^2] \\
 &= \text{Var}[h(\mathbf{x})] + \text{Bias}[h(\mathbf{x})]^2 + E_P[\varepsilon^2] \\
 &= \text{Var}[h(\mathbf{x})] + \text{Bias}[h(\mathbf{x})]^2 + \sigma^2
 \end{aligned}$$

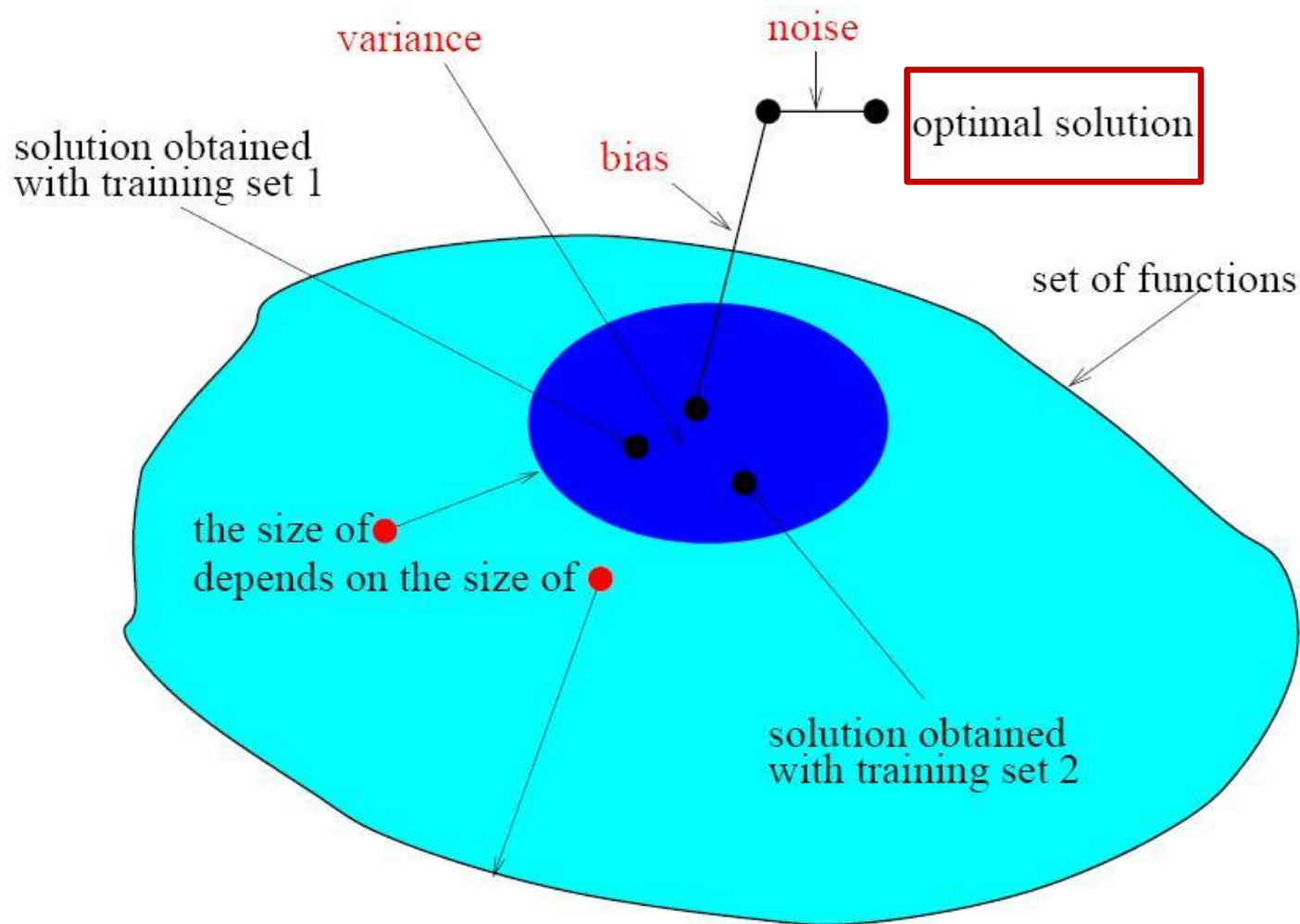
$E_P [h(\mathbf{x})^2] + E_P [y^2] - 2E_P[y]E_P[h(\mathbf{x})]$

- Expected prediction error = Variance + Bias<sup>2</sup> + Noise<sup>2</sup>

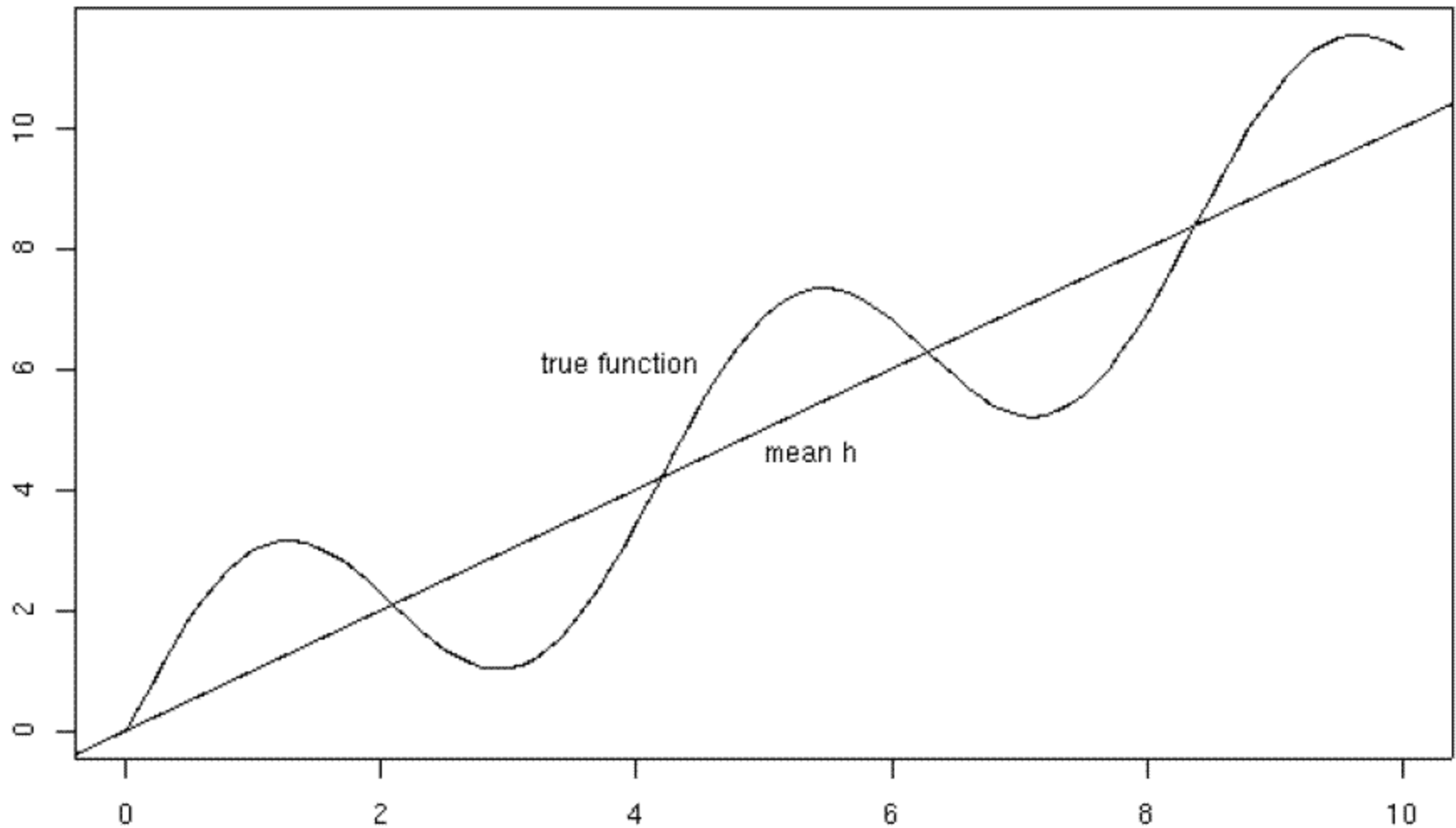
# Bias/Variance results

- Decomposition of expected (on various TR) error at a point  $\mathbf{x}$ :
  - **Bias**: quantifies the discrepancy between true function and  $h(\mathbf{x})$  (with  $h(\mathbf{x})$  averaged over different TR data) [*systematic error*]
    - E.g. Due to too small  $H$ , too rigid model  $[\bar{h}(\mathbf{x}) - f(\mathbf{x})]^2$
  - **Variance**: quantifies the variability of the response of model  $h$  for different realizations of the training data
    - Due to too high flexibility  $E_P [(h(\mathbf{x}) - \bar{h}(\mathbf{x}))^2]$
  - **Noise**: even the optimal solution could be wrong!
    - E.g. if for a given  $\mathbf{x}$  there are more than one possible  $d$   
[*irreducible*: does not depends on the model,  
we have a tolerance on the response ( $\sigma$ , sigma)]

# Bias-Variance (graphical view)

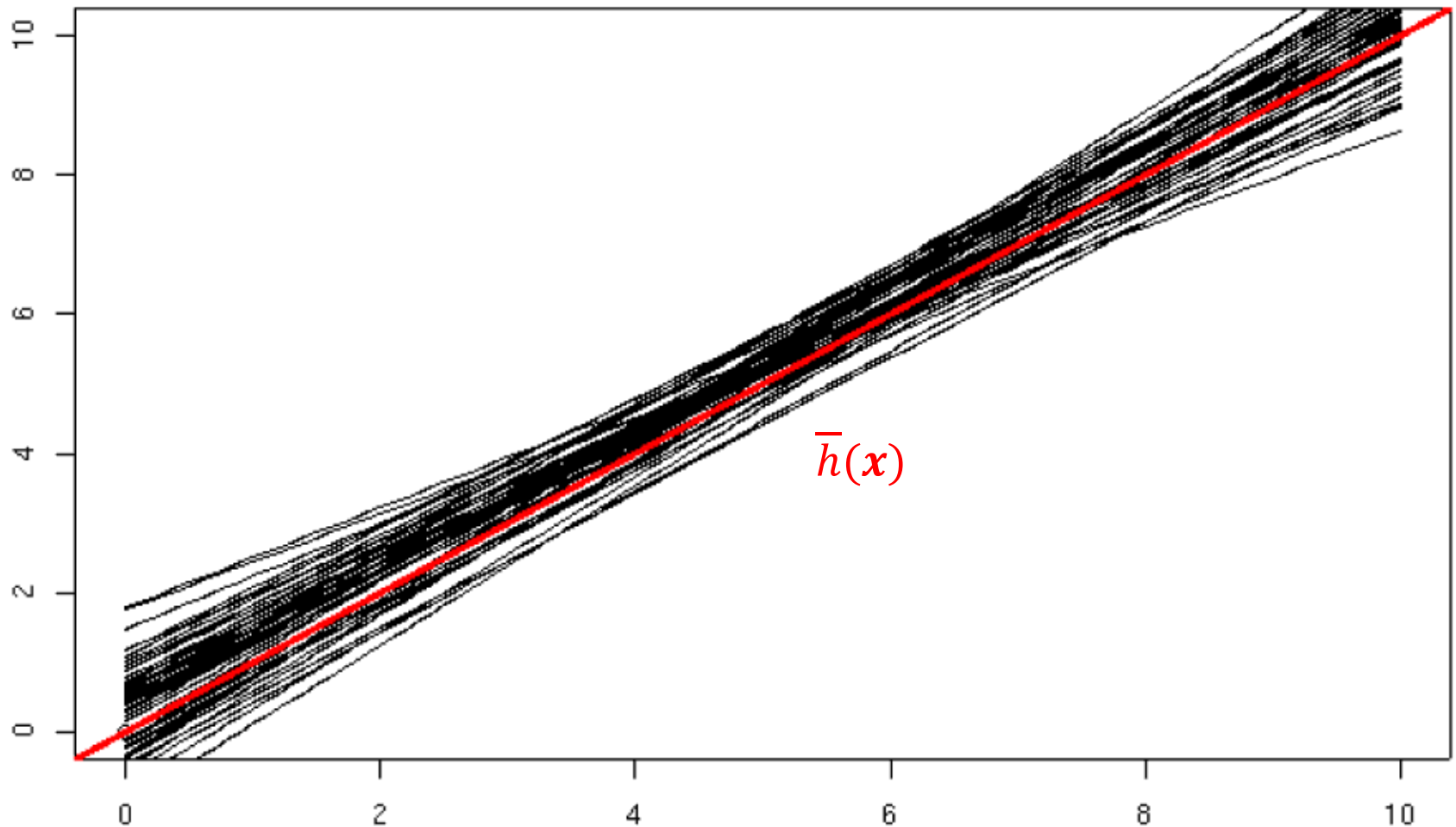


# Bias

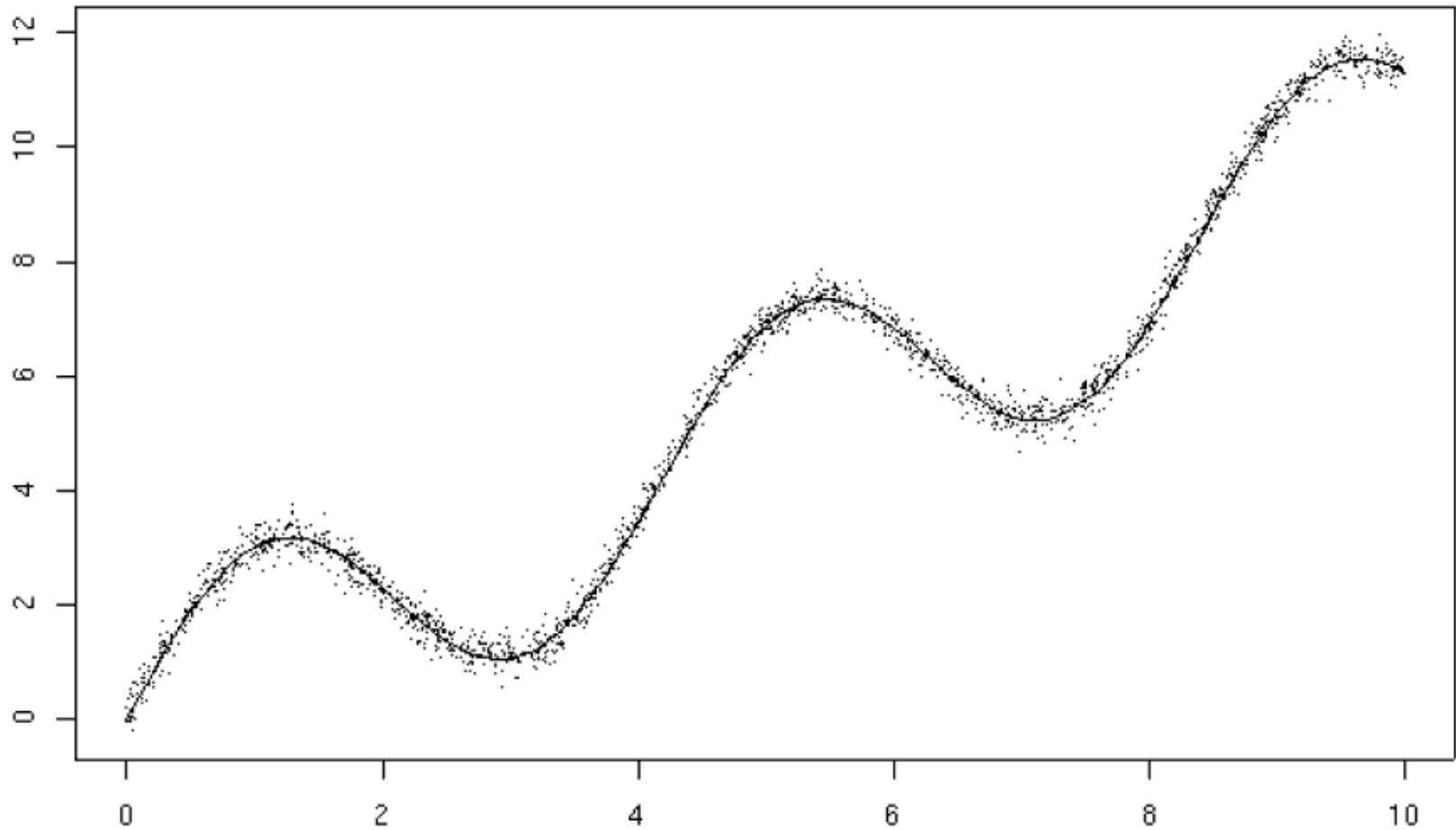


Note the high bias ! (the linear model is too rigid)

# Variance

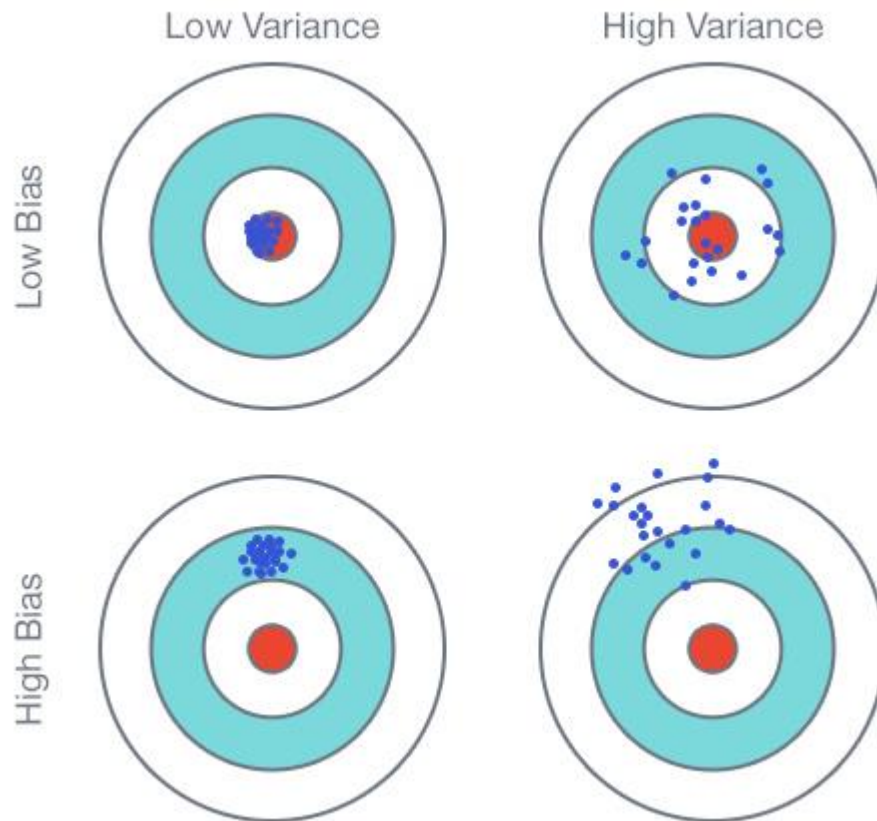


# Noise





# A nice graphical analogy for bias and variance



Darts game

Exercise: identify under and overfitting cases...  
But the match of the analogy is not perfect for ML...

# Bias/Var and Regularization

Recall Regularization:

$$\text{Loss}(\mathbf{w}) = \sum_p (d_p - o(\mathbf{x}_p))^2 + \lambda \|\mathbf{w}\|^2$$

Left (data) term (TR data)    Penalty term

Varying  $\lambda$  we can obtain

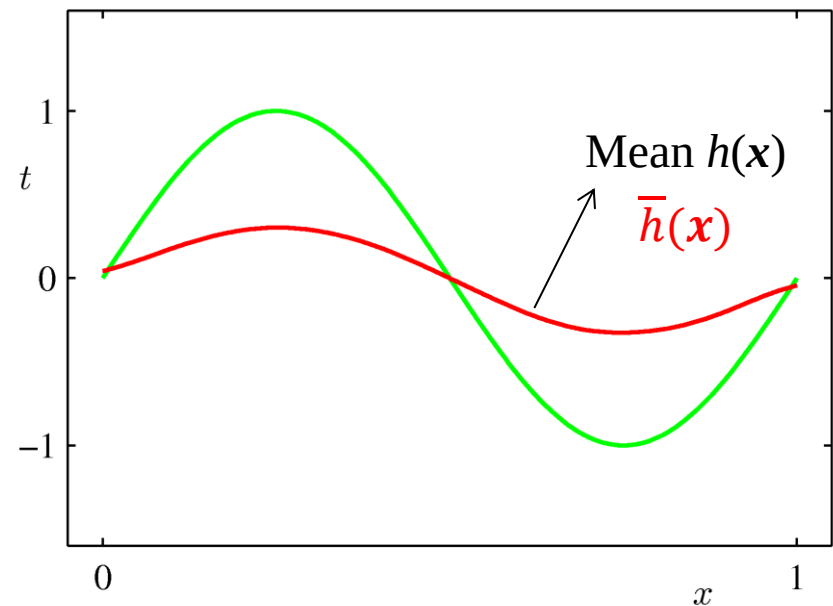
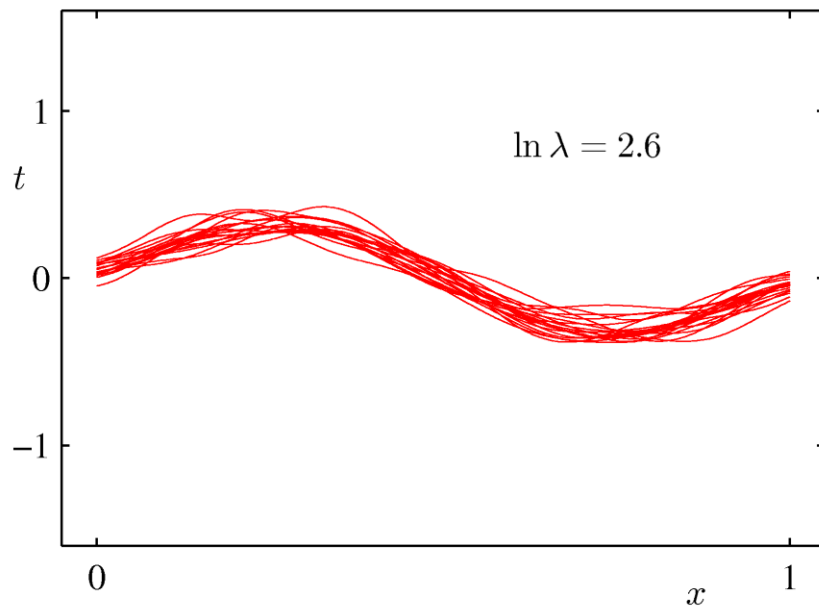
- less regularized (complex) solutions (low  $\lambda$ )
- more regularized (less complex) solutions (high  $\lambda$ )

- The following example (from Bishop) (should be) using RBF network with LMS

(By the way, note: that the intercept is typically left out of the penalty term)

# The Bias-Variance Decomposition: high lambda

- Example: 25 data sets from the sinusoidal, varying the degree of regularization, lambda ( $\lambda$ )



High bias, low variance

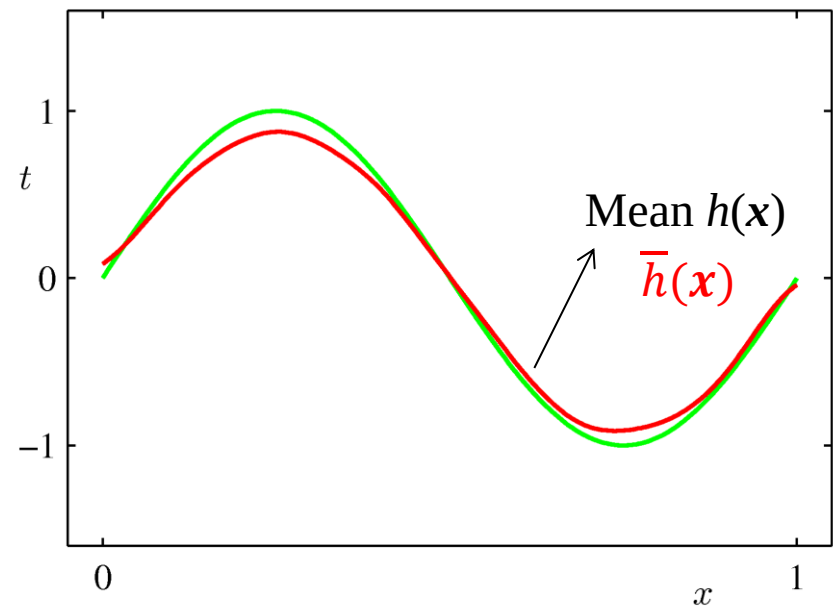
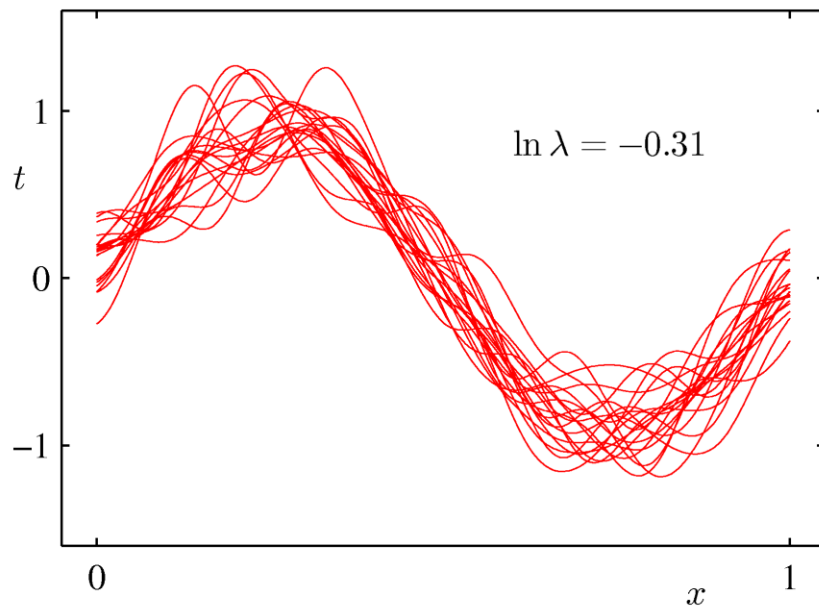
High  $\lambda$  increase the effect of “penalty term”,  
less dependent on left (data) term (“too rigid”)

# The Bias-Variance Decomposition: medium lambda



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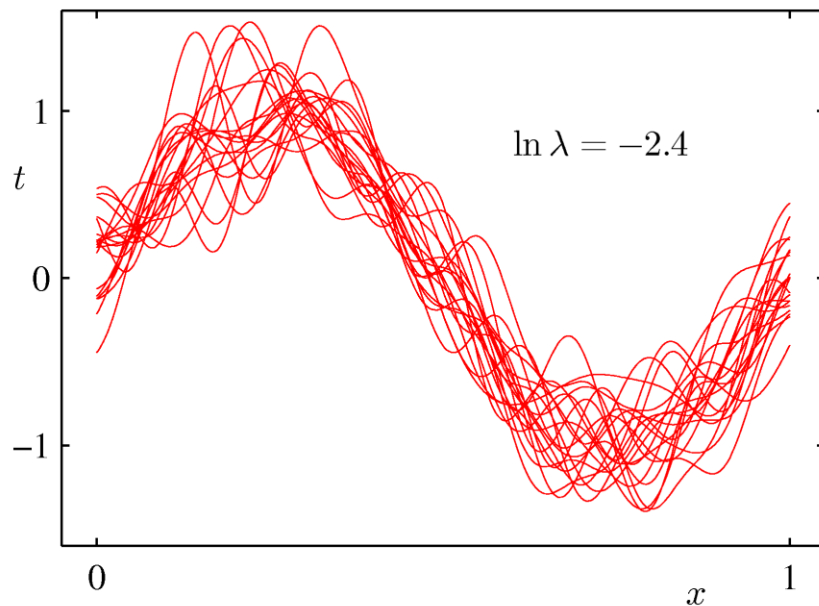
- Example: 25 data sets from the sinusoidal, varying the degree of regularization, lambda ( $\lambda$ ).



Lower bias, higher variance  
Decreasing  $\lambda \rightarrow$  we are more dependent  
on the specific training data (left –data- term)

# The Bias-Variance Decomposition: low lambda

- Example: 25 data sets from the sinusoidal, varying the degree of regularization, lambda ( $\lambda$ ).



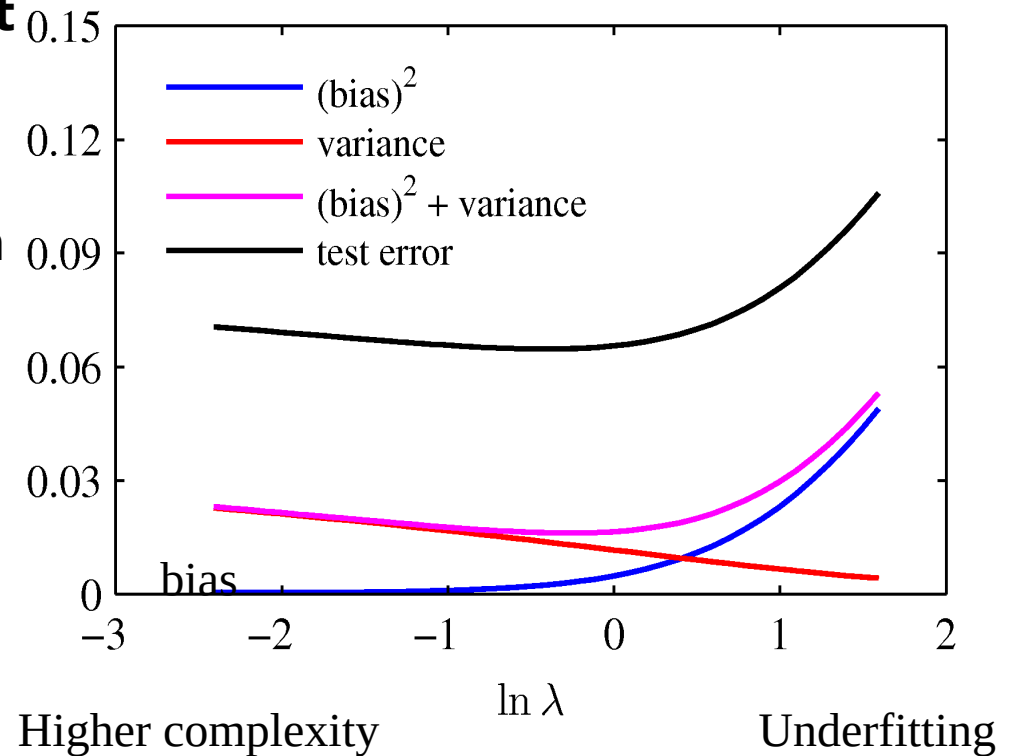
Very low  $\lambda$ : Low bias, high variance:

Test error (for a given TR) can be high (the variance component) but ... even if  $h(x)$  are quite variegated, they are correct “on average” (right plot)

# The Bias-Variance Trade-off

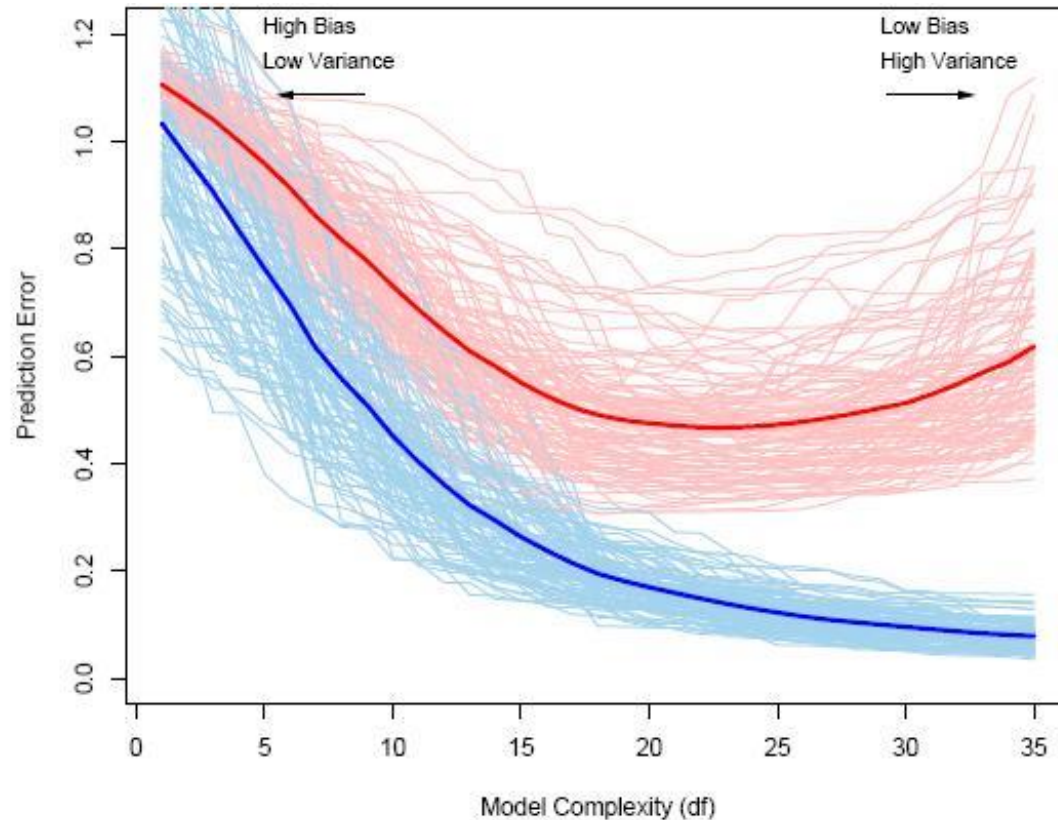
- From these plots, we note that an over-regularized model (large  $\lambda$ ) will have a high bias, while an under-regularized model (small  $\lambda$ ) will have a high variance.

- Again the trade-off!



Note: complexity is higher on the left  
(reverse of our standard plots)

# Bias-Variance plot\* (with TR and TS)



unlucky

lucky

Complex model : lower  
bias, higher variance →  
possible overfitting

**FIGURE 7.1.** Behavior of test sample and training sample error as the model complexity is varied. The light blue curves show the training error  $\overline{err}$ , while the light red curves show the conditional test error  $Err_T$  for 100 training sets of size 50 each, as the model complexity is increased. The solid curves show the expected test error  $Err$  and the expected training error  $E[\overline{err}]$ .

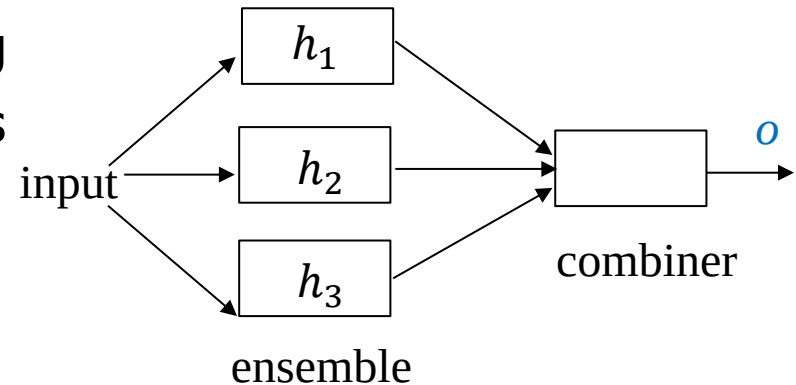
# Other ML topics

- Ensemble learning: bagging, boosting
- Feature selection
- PAC learning
  - E.g. Mitchell, Haykin
- Comparative analysis of inductive principia (SRM, MDL, Bayesian inference, Regularization) with respect to complexity control
  - Cherkassky, Mulier (chapter 2)



# Ensemble Learning

- A first look to **ensemble learning**
- Take advantage of multiple models



Simplest case (“voting schema”):

- Regression
  - simple average committee  $o(\mathbf{x}) = 1/K \sum_{i=1}^K h_i(\mathbf{x}) \Rightarrow \bar{h}(\mathbf{x})$
  - $loss(E_i(h_i)) \leq E_i(loss(h_i))$  I.e.
  - (mean<sub>D</sub>) square error of a committee  $\leq$  mean of the (mean<sub>D</sub>) square error  
 e.g.  $(t-o)^2$ ,  $t=5$ , **2 models**  $\rightarrow ((5-2)^2 + (5-4)^2)/2 = 5 > (5-3)^2 = 4$   
 or even:  $((5-6)^2 + (5-4)^2)/2 = 1$  while  $(5-5)^2 = 0$
- Classification
  - Take a vote over many classifier (given they provide “diversified” responses): why ? (or classify after the mean of the continuous outputs)

Alternative (“stacking schema”): the combiner is again a ML model

# Bagging (*bootstrap aggregating*)

- Combine many classifiers
- Bagging
  - Train  $K$  classifiers on different subsets of training set
  - Differentiate each training using bootstrap (resampling with replacement)
- Think to bias-variance (*see "low lambda" case slide: right figure*):  $\bar{h}(x)$
- High variance models (with low bias) can perform well *on average*
- Or in other terms: the average of  $h$  reduces the variance
  - Regression: use the mean
  - Classification: voting of  $K$  classifiers
    - Likely to increase **margin** of a classifier
    - Indeed, average of  $K$  separation curves likely to be "in the middle"  
"In medio stat virtus"

# Boosting (e.g. AdaBoost)

- If models have the same errors we have not advantage on ensembling
  1. Differentiate each training concentrating on errors (more weight to difficult instances, e.g. more likely to be included in the TR of the next classifier) E.g.:
    - Train 1st classifier
    - Train 2nd classifier, weight more instances misclassified before
    - Etc.
  2. Combine results by output weights (weighted vote for classifiers, with more weight to low error classifier)
- If not stopped, boosting will learn to classify correctly all instances from training set (using *weak learners* that can merely do better than random guessing, i.e. error less than  $\frac{1}{2}$  for binary classification) providing an incremental approach to construct complex models
- Can greatly improve performance of *weak learners*
- Even if bias is reduced the variance does not increase (resist to overfitting)
- As bagging, also likely to produce separation curves *in the middle* (i.e. max margin)
- But can suffer for noisy data (which are weighted more)! (it does not solve the underfitting/overfitting issue in itself)

# Feature Selection (few words)

Find a *selection of the features* that are more informative for the problem at hand....

Great benefits for

- Dimensionality reduction (e.g. VC-dim effects for NN, curse of dim, ...), filtering of irrelevant information and noise  
(that can lead to great performance improvement)
- Interpretability (e.g. find a set of useful descriptors)
- ❖ It is computationally hard (think how many possible subsets of features can be composed for the learner, and consider the retraining of models)
- ❖ Typically, a heuristic search of the best subset (including or not the learner) by greedy or GA or other optimization techniques.
- ❖ Results are particularly sensible to the validation method: "*feature subset selection bias*", wrong use if selection is made on the entire data set (see slides on CV!)

A survey: I. Guyon, A. Elisseeff. "An Introduction to Variable and Feature Selection". JMLR vol.3:1157-1182, 2003

# Bibliography (on bias-variance)



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- Haykin (2<sup>nd</sup> ed.) sec. 2.13:pp.84-89 : they are not complete !!!
- Haykin (3<sup>rd</sup> ed.) sec. 2.7:pp.112-116: they are not complete !!!
- Cherkassky, Mulier: 3.4.3
- Hastie, Tibshirani, Friedman : cap 1 e 7

Also ...

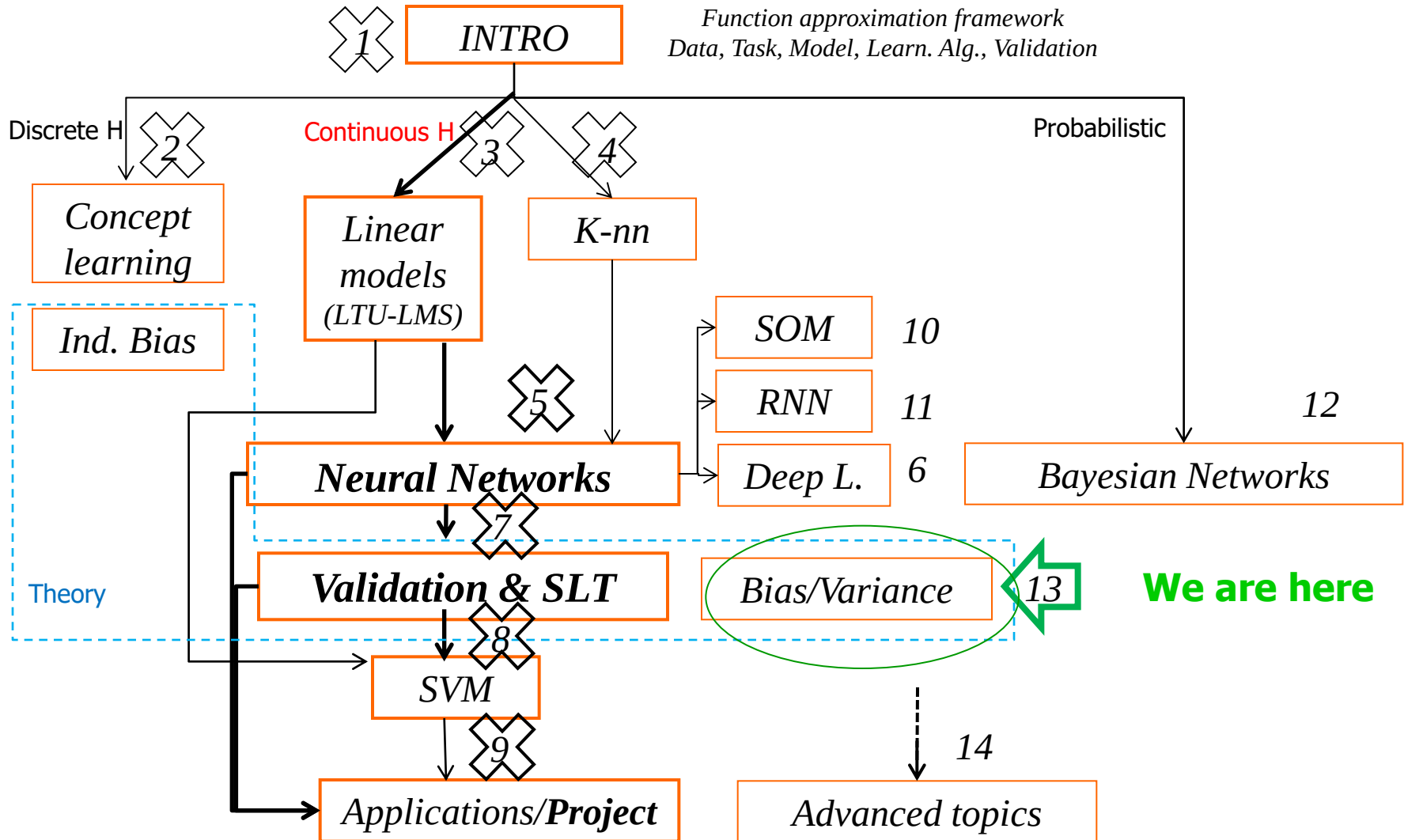
- Bishop,
- Duda, Hart, Stork

# ML Course structure

## Where we are and go



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